

Persistent Branches and Transcritical Exchange

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Definition: Persistent equilibrium branch

Let $R \subseteq \mathbb{R}$ be a specified parameter set. For $\dot{x} = f(x; r)$, a **persistent equilibrium branch on R** is a continuous function $x^* : R \rightarrow \mathbb{R}$ satisfying

$$f(x^*(r); r) = 0 \quad \text{for every } r \in R.$$

Definition: Transcritical bifurcation

A **transcritical bifurcation** occurs when two equilibrium branches cross and exchange stability.

Definition: Transcritical normal form

The **transcritical normal form** is

$$\dot{x} = rx - x^2.$$

Question: How does the transcritical normal form exchange stability?

Analyze

$$\dot{x} = rx - x^2 = x(r - x).$$

Solution. The equilibrium branches are

$$x_0^*(r) = 0, \quad x_1^*(r) = r.$$

Since

$$f_x(x; r) = r - 2x,$$

the branch rates are

$$\lambda_0(r) = f_x(0; r) = r, \quad \lambda_1(r) = f_x(r; r) = -r.$$

Therefore

	$x^* = 0$	$x^* = r$
$r < 0$	asymptotically stable	unstable
$r > 0$	unstable	asymptotically stable

At $r = 0$, $\dot{x} = -x^2 \leq 0$, so the origin repels from the left and attracts from the right; it is half-stable.

Visualization: Transcritical branch crossing

r	ordered equilibria	phase line
$r < 0$	$r < 0$	$\leftarrow \circ_{x=r} \rightarrow \bullet_{x=0} \leftarrow$
$r = 0$	0	$\leftarrow \odot_{x=0} \leftarrow$
$r > 0$	$0 < r$	$\leftarrow \circ_{x=0} \rightarrow \bullet_{x=r} \leftarrow$

The bifurcation diagram is the crossing of

$$x^* = 0 \quad \text{and} \quad x^* = r,$$

with solid and dashed segments exchanged at $(r, x^*) = (0, 0)$.

Recall: Scalar implicit function theorem

Let g be C^1 near $(0, 0)$, with $g(0; 0) = 0$ and $g_x(0; 0) \neq 0$. Near $(0, 0)$, the equation $g(x; r) = 0$ defines a unique C^1 function $x = \chi(r)$ satisfying

$$\chi(0) = 0, \quad \chi'(0) = -\frac{g_r(0; 0)}{g_x(0; 0)}.$$

Lemma: Factorization forced by a persistent zero branch

Let $f \in C^2$ near $(0, 0)$ and suppose $f(0; r) = 0$ for all sufficiently small r . Then

$$f(x; r) = xg(x; r)$$

for a C^1 function g .

Proof. Define

$$g(x; r) = \int_0^1 f_x(sx; r) ds.$$

Since $f \in C^2$, differentiation under the integral shows that $g \in C^1$. The fundamental theorem of calculus gives

$$xg(x; r) = \int_0^x f_x(u; r) du = f(x; r) - f(0; r) = f(x; r).$$

□

Proposition: Exchange of stability on two local branches

For the parameterized system $\dot{x} = f(x; r)$, let $f(x; r) = xg(x; r)$, where $g \in C^1$ near $(0, 0)$, and suppose

$$g(0; 0) = 0, \quad a = g_r(0; 0) \neq 0, \quad b = g_x(0; 0) \neq 0.$$

Then the local equilibrium branches and their linearized rates are

$$x_0(r) = 0, \quad x_1(r) = -\frac{a}{b}r + o(r) \quad (r \rightarrow 0),$$

$$\lambda_0(r) = ar + o(r), \quad \lambda_1(r) = -ar + o(r) \quad (r \rightarrow 0).$$

Hence the branches exchange stability at $r = 0$.

Proposition: Exchange of stability on two local branches (continued)**Proof.** Since

$$g(x; r) = ar + bx + o(|r| + |x|) \quad ((x, r) \rightarrow (0, 0)),$$

the implicit function theorem gives

$$g(x_1(r); r) = 0, \quad x_1(r) = -\frac{a}{b}r + o(r) \quad (r \rightarrow 0).$$

On $x_0(r) = 0$,

$$\lambda_0(r) = f_x(0; r) = g(0; r) = ar + o(r) \quad (r \rightarrow 0).$$

On the nonzero branch, $g(x_1(r); r) = 0$, so

$$\lambda_1(r) = x_1(r)g_x(x_1(r); r) = -ar + o(r) \quad (r \rightarrow 0).$$

The signs are opposite for all sufficiently small $r \neq 0$. □**Question: Where is the bifurcation curve in a two-parameter model?**For real parameters a, b , consider

$$\dot{x} = x(1 - x^2) - a(1 - e^{-bx}),$$

and find the transcritical bifurcation curve at $x = 0$ and the leading nonzero equilibrium branch.**Solution.** The origin is an equilibrium for every (a, b) . Fix any $b \neq 0$ and vary a near $1/b$. Taylor expansion as $x \rightarrow 0$, uniformly for these a , gives

$$1 - e^{-bx} = bx - \frac{1}{2}b^2x^2 + O(x^3),$$

so

$$\dot{x} = (1 - ab)x + \frac{1}{2}ab^2x^2 + O(x^3).$$

The linear coefficient vanishes on

$$\boxed{ab = 1}.$$

For $x^* \neq 0$, division by x^* gives

$$0 = 1 - ab + \frac{1}{2}ab^2x^* + O((x^*)^2).$$

At $(x^*, a) = (0, 1/b)$, the derivative of this expression with respect to x^* is $b/2 \neq 0$. The implicit function theorem therefore gives the second branch, and the displayed equation gives its expansion:

$$x^* = \frac{2(ab - 1)}{ab^2} + O((ab - 1)^2) \quad (a \rightarrow 1/b).$$

Since $f_x(x; a, b) = 1 - ab + ab^2x + O(x^2)$, substitution gives the branch rates

$$\lambda_0 = 1 - ab, \quad \lambda_1 = ab - 1 + O((ab - 1)^2) \quad (a \rightarrow 1/b).$$

These have opposite signs for all sufficiently nearby parameters with $ab \neq 1$.

Question: How is a nonzero persistent branch moved to the origin?

Analyze

$$\dot{x} = r \log x + x - 1, \quad x > 0,$$

near the persistent equilibrium $x = 1$.

Solution. Set

$$u = x - 1.$$

Taylor expansion as $u \rightarrow 0$, uniformly for r near -1 , gives

$$\log(1 + u) = u - \frac{1}{2}u^2 + O(u^3),$$

so

$$\dot{u} = (r + 1)u - \frac{r}{2}u^2 + O(u^3).$$

The bifurcation value is

$$r_c = -1.$$

For a nonzero equilibrium, divide by u :

$$0 = (r + 1) - \frac{r}{2}u + O(u^2).$$

The derivative with respect to u at $(u, r) = (0, -1)$ is $1/2 \neq 0$, so the implicit function theorem gives a second local branch. Solving to leading order and substituting into $f_x(x; r) = r/x + 1$ gives

$$\begin{aligned} x_0^* &= 1, & \lambda_0 &= r + 1, \\ x_1^* &= 1 + \frac{2(r + 1)}{r} + O((r + 1)^2), & \lambda_1 &= -(r + 1) + O((r + 1)^2), \end{aligned} \quad (r \rightarrow -1).$$

For $r < -1$ sufficiently close to -1 , x_0^* is asymptotically stable and x_1^* is unstable; for $r > -1$, their stability is reversed. At $r = -1$,

$$\dot{u} = \frac{1}{2}u^2 + O(u^3) > 0 \quad (0 < |u| \text{ sufficiently small}),$$

so $x = 1$ attracts from the left and repels to the right: it is half-stable.

Question: Is a rational vector field still transcritical?

Analyze

$$\dot{x} = \frac{x(r + x)}{1 + x^2}.$$

Solution. The equilibrium branches are

$$x_0^* = 0, \quad x_1^* = -r.$$

At either equilibrium, the numerator vanishes, so differentiation gives

$$f_x(x^*; r) = \frac{r + 2x^*}{1 + (x^*)^2}.$$

Question: Is a rational vector field still transcritical? (continued)

Therefore

$$f_x(0; r) = r, \quad f_x(-r; r) = -\frac{r}{1 + r^2}.$$

For $r < 0$, $x^* = 0$ is asymptotically stable and $x^* = -r$ is unstable; for $r > 0$, their stability is reversed. At $r = 0$,

$$\dot{x} = \frac{x^2}{1 + x^2} \geq 0,$$

so the origin attracts from the left and repels to the right: it is half-stable. Near $(x, r) = (0, 0)$,

$$\dot{x} = rx + x^2 + O(|x|^3 + |r|x^2) \quad ((x, r) \rightarrow (0, 0)).$$

The state reversal $y = -x$ gives the leading form

$$\dot{y} = ry - y^2.$$

Idea: Mathematical and physical branches

If x denotes a population, concentration, or intensity, the physical state space may be $x \geq 0$. A branch with $x^* < 0$ remains part of the mathematical bifurcation diagram but is not a physically admissible state.